

Template for Research Methodology

BY

Student Name

ID: 1464242

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of B.Sc.(Engg.) or
M.Sc.(Engg.) in Information and Communication Technology



Department of Information and Communication Technology (ICT)

Mawlana Bhashani Science and Technology University

Santosh, Tangail-1902, Bangladesh

2025

Declaration

I hereby declare that this thesis is my own original work and has not been submitted previously for any degree at this or any other institution.

Student Name

Date: _____

Acknowledgements

I would like to express my sincere gratitude to ...

Abstract

This thesis investigates ...

Contents

1	Introduction	8
1.1	Background of the Study	8
1.2	Problem Statement	8
1.3	Motivation	8
1.4	Research Questions and Objectives	8
1.5	Scope of the Study	9
1.6	Significance of the Study	9
1.7	Thesis Organization	9
2	Literature Review	10
2.1	Introduction	10
2.2	Theoretical Background	10
2.3	Related Work	10
2.4	Research Gap	10
2.5	Summary	11
3	Methodology	12
3.1	Research Framework	12
3.2	System Architecture	12
3.3	Materials, Tools, and Software	12
3.4	Dataset Description	12
3.5	Preprocessing and Feature Engineering	13

3.6	Proposed Algorithms / Models	13
3.7	Experimental Setup	13
3.8	Evaluation Metrics	13
3.9	Summary	13
4	Results and Analysis	14
4.1	Introduction	14
4.2	Experimental Results	14
4.3	Analysis of Results	14
4.4	Comparison with Existing Methods	14
4.5	Discussion of Limitations	15
4.6	Summary	15
5	Discussion	16
5.1	Interpretation of Findings	16
5.2	Implications	16
5.3	Potential Applications	16
5.4	Summary	16
6	Conclusion and Future Work	17
6.1	Summary of the Research	17
6.2	Key Contributions	17
6.3	Conclusion	17
6.4	Limitations	17
6.5	Future Work	18
A	Additional Data	20
B	Source Code	21

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background of the Study

Write your background here.

1.2 Problem Statement

Clearly state the problem your thesis addresses.

1.3 Motivation

Explain why this problem is important.

1.4 Research Questions and Objectives

List your research questions and specific objectives.

1.5 Scope of the Study

Describe what is included and excluded.

1.6 Significance of the Study

Explain who benefits from this research.

1.7 Thesis Organization

Briefly describe what each chapter contains.

Chapter 2

Literature Review

2.1 Introduction

Short intro to the chapter.

2.2 Theoretical Background

Relevant theory, definitions, and models.

2.3 Related Work

Discuss existing methods and prior research.

2.4 Research Gap

Show what is missing in the literature.

2.5 Summary

Short wrap-up of the chapter.

Chapter 3

Methodology

3.1 Research Framework

Give a high-level view (flowchart, block diagram).

3.2 System Architecture

Describe the system/model architecture.

3.3 Materials, Tools, and Software

List sensors, equipment, Python libraries, etc.

3.4 Dataset Description

Explain data collection or source.

3.5 Preprocessing and Feature Engineering

Describe cleaning, normalization, feature extraction.

3.6 Proposed Algorithms / Models

Explain your algorithm, equations, and logic.

3.7 Experimental Setup

Hardware, software environment, parameter settings.

3.8 Evaluation Metrics

MAE, RMSE, Accuracy, F1-score, etc.

3.9 Summary

Short recap of methodology.

Chapter 4

Results and Analysis

4.1 Introduction

Intro to what you present here.

4.2 Experimental Results

Present tables, graphs, figures.

4.3 Analysis of Results

Explain what the results mean.

4.4 Comparison with Existing Methods

Compare with baseline models or literature.

4.5 Discussion of Limitations

Explain weaknesses and constraints.

4.6 Summary

Wrap-up of the findings.

Chapter 5

Discussion

5.1 Interpretation of Findings

Connect results to research questions.

5.2 Implications

Practical and theoretical implications.

5.3 Potential Applications

Where your method can be used (e.g., sensors, healthcare).

5.4 Summary

Short summary of discussion.

Chapter 6

Conclusion and Future Work

6.1 Summary of the Research

Summarize what was done.

6.2 Key Contributions

List the main contributions of your thesis.

6.3 Conclusion

Final take-away messages.

6.4 Limitations

State the limitations honestly.

6.5 Future Work

Propose what can be done next.

Bibliography

- [1] Author, A. B., & Author, C. D. (Year). *Title of the paper or book*. Journal/Publisher.
- [2] Another reference example.

Appendix A

Additional Data

Put raw data, extra tables, etc.

Appendix B

Source Code

List important code snippets or link to repository.